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Bayesian Port–Hamiltonian Surrogate for Three-Phase Reservoir Flow Simulation

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Abstract

We present a learned structure-preserving surrogate operator in the form of a Hamiltonian Network that reproduces full-physics three-phase slightly-compressible flow simulations at roughly 10× lower computational cost while retaining fundamental conservation laws. We express the reservoir dynamics as a finite-volume *port–Hamiltonian* (pH) dynamic system and to progressively learn the Hamiltonian through its dynamic vector field with Bayesian optimization, and using a scalable mixture-of-Gaussian-process (MoGP) prior. The pH scaffold enforces mass balance, passivity, and Lyapunov stability by construction; the Bayesian GP treatment supplies calibrated epistemic uncertainty that propagates through the surrogate estimation state process and can be used to quantify forecast uncertainty in closed-loop reservoir management. Unlike previously published neural-operator and reduced-order surrogates, our method guarantees conservation without ad-hoc penalties and delivers analytic gradients for adjoint-free history matching.

Introduction

Three-phase finite-volume reservoir simulators underpin forecasting, history matching, and closed-loop optimization, yet a single run still requires minutes to hours on modern GPUs. A closed-loop workflow demands 10^4 – 10^5 forward evaluations (Nasir and Durlofsky 2023), rendering exhaustive reliance on full-physics codes prohibitive. Projection-based reduced-order models (ROMs) shorten runtimes but lose accuracy once well controls depart from the training trajectory (Cardoso and Durlofsky 2010a;b). Purely data-driven surrogates – convolutional U-Nets (Tang et al. 2020), Fourier Neural Operators (FNOs) (Li et al. 2020; Wen et al. 2022; Badawi and Gildin 2025), and DeepONets (Lu et al. 2021) – retain accuracy under new controls but do not enforce conservation, so mass and energy drift during long rollouts. Structure-preserving networks, including physics-informed neural networks (PINNs), theory-guided CNNs, and port–Hamiltonian NNs (Roth et al. 2025; Moradi et al. 2025), embed PDE residuals or passivity constraints, yet published reservoir-flow demonstrations remain limited to 2-D grids with (10^2-10^3) cells, focus on singleor two-phase physics, and rarely quantify uncertainty – typically reporting only parameter-level variances rather than calibrated predictive intervals (Zhu and Zabaras 2018; Wang et al. 2022). A three-phase surrogate that is both conservative and uncertainty-aware has not been reported. We target precisely this gap, restricting attention to three-phase, slightly compressible flow without dissolution or vaporization; extending the port–Hamiltonian surrogate to full black-oil PVT is deferred to future work.

We construct a surrogate that combines

i. a **port–Hamiltonian scaffold** that rewrites the finite-volume three-phase, slightly compressible Darcy equations in energy form, thereby conserving mass and supplying a Lyapunov function; and
ii. **Bayesian learning of the Hamiltonian**, which models the cell-wise energy density with a mixture of stationary Gaussian processes approximated by random Fourier features, providing analytic means, variances, and gradients.

The resulting surrogate (i) preserves the governing structure, (ii) is fully differentiable, and (iii) yields calibrated epistemic uncertainty.

Contributions. We introduce a port–Hamiltonian formulation of a three-phase, slightly compressible flow that explicitly separates interconnection, dissipation, and well ports. The Hamiltonian is parameterized with a Bayesian mixture of Gaussian processes. The passivity-preserving Crank–Nicolson integrator ensures mass conservation and bounded energy drift.

Paper outline. The Background section reviews port–Hamiltonian systems and Gaussian-process regression. The Methodology section details the surrogate formulation and training procedure. The Experiments section evaluates performance on the SPE1 benchmark. The Related Work section compares our approach with existing ROMs, neural operators, and physics-informed surrogates. The Conclusion summarizes key findings and future directions.

Background

Three-Phase slightly-compressible Flow

Finite-volume simulators solve the three-phase mass-balance equations (Rasmussen et al. 2021)

$$\partial_t \phi S_\alpha \rho_\alpha + \nabla \cdot \rho_\alpha \mathbf{u}_\alpha = q_\alpha, \quad \mathbf{u}_\alpha = -\frac{k k_{r\alpha}}{\mu_\alpha} \nabla p - \rho_\alpha g \nabla z, \quad \alpha \in \{o, w, g\}. \tag{1}$$

Throughout, we neglect dissolution/vaporization (no R_s, R_v) and assume slightly-compressible PVT, $\rho_\alpha(p) = \rho_\alpha^{\mathrm{ref}}\left[1 + c_\alpha\left(p - p_{\mathrm{ref}}\right)\right]$. The first term is the rate of change of phase mass per control volume, the second the divergence of mass flux, and q_α the well or source term. Velocity $\mathbf{u}_\alpha$ follows Darcy's law with permeability k, relative permeability $k_{r\alpha}$, viscosity μ_α, the common pressure gradient ∇p(capillary pressures neglected), gravity g, and elevation gradient∇z. Phase behavior closes the system; capillary pressures are neglected and each phase is treated as slightly compressible, $\rho_\alpha(p) = \rho_\alpha^{\mathrm{ref}}\ 1 + c_\alpha(p - p_{\mathrm{ref}})$ (Aziz and Settari 1979).

The saturations satisfy the algebraic closure condition $S_o + S_w + S_g = 1$. In the energy-based surrogate developed later we therefore keep the three extensive phase masses $m_\alpha = \phi V \rho_\alpha S_\alpha$ as explicit state variables, so the reduced state reads $x = [m_o, m_w, m_g]^\top$.

At every function evaluation the common pressure p is recovered algebraically by solving the scalar nonlinear equation

$$g(p) := \frac{m_o}{\rho_o(p)} + \frac{m_w}{\rho_w(p)} + \frac{m_g}{\rho_g(p)} - \phi V = 0.$$

Because $\rho_\alpha \approx \rho_\alpha^{\mathrm{ref}}\, 1 + c_\alpha \Delta p$, the residual $g(p)$ is smooth and a small fixed number (two to three) of Newton iterations suffices to reduce $|g(p)|$ below $10^{-8}\phi V$ in practice. With the converged pressure we set

$$S_\alpha = \frac{m_\alpha}{\phi V \rho_\alpha(p)}, \quad \alpha \in \{o, w, g\},$$

so that the closure $S_o + S_w + S_g = 1$ holds identically.

Stiffness and non-linearity still require Newton–Krylov iterations in classical simulators, which increases runtime; the surrogate presented later aims to cut that cost without loss of accuracy.

Port–Hamiltonian Systems

A finite-dimensional port–Hamiltonian (pH) system reads

$$\dot{x} = J(x) - R(x)\nabla_x H(x) + B(x)u, \quad y = B^\top(x)\nabla_x H(x), \tag{2}$$

where $x \in \mathbb{R}^n$ contains the extensive storage variables, H is the Hamiltonian, and (u, y) are the powerconjugate port variables (van der Schaft and Jeltsema 2014). The interconnection matrix satisfies $J = -J^\top$; the dissipation matrix obeys $R \geq 0$. Passivity gives $\dot{H} = -\nabla_x^\top H\, R\, \nabla_x H + u^\top y \leq u^\top y$, so the stored energy cannot increase when $u \equiv 0$.

Mapping the slightly compressible flow discretization to the pH form. Following the Dirac structure, we retain only the extensive phase masses as dynamic variables. Pressure and saturations are reconstructed algebraically, ensuring that the closure condition $S_o + S_w + S_g = 1$ is automatically satisfied. With N cells we define

$$x = m_{o,1}, \ldots, m_{o,N}, \; m_{w,1} \ldots, m_{w,N}, \; m_{g,1}, \; \ldots, m_{g,N}{}^\top \in \mathbb{R}^{3N}.$$

We collect the discrete divergence over all faces and phases in $G \in \mathbb{R}^{3N \times Ne}$ and set the dissipation matrix

$$R = G M_\theta G^\top \succcurlyeq 0,$$

where M_θ is the block-diagonal mobility–permeability matrix.

The port matrix $B \in \mathbb{R}^{3N \times nw}$ has entries $B_{(\alpha,i),w} = \rho_{\alpha,i} T_{w,i} M_{\alpha,i}$, $\quad \alpha \in \{o, w, g\}$, $\alpha \in \{o, w, g\}$. We freeze $B\, x_k$ over $[t_k, t_{k+1}]$; treating R and B as piecewise constant introduces a passivity defect of order $O(\Delta t^3)$, which vanishes as $\Delta t \to 0$ and was found negligible for the time steps used in the experiments.

The cell-wise Hamiltonian density $h_\theta(m_o, m_w, m_g; \kappa, \phi)$ is learned with a Bayesian mixture of stationary Gaussian processes. Automatic differentiation provides $_x H_\theta$ together with analytic mean and variance estimates. We shift h_θ by its thermodynamic baseline so that $H_\theta(x) \geq 0$ for all admissible states, preserving the Lyapunov property.

Methodology

State, static features & control

Static features. Rock properties are stored on an N–cell Cartesian grid in two cell-wise tensors: $\kappa \in \mathbb{R}^{N\times 3}$ for principal permeabilities and $\phi \in \mathbb{R}^{N\times 1}$ for porosity. They are concatenated and standard-scaled as

$$z^{\mathrm{stat}} = \left[\log_{10} k_x, \log_{10} k_y, \log_{10} k_z, \phi\right].$$

Dynamic snapshot. At simulator step k the surrogate state contains the three extensive phase masses

$$x_k = \left[m_o, m_w, m_g\right] \in \mathbb{R}^{3N},$$

with $m_{\alpha,i} = \phi_i V_i \rho_\alpha(p_i) S_{\alpha,i}$ for $\alpha \in \{o, w, g\}$. Pressure is reconstructed from the scalar balance equation $g_i(p) = 0$, after which the algebraic closure $S_o + S_w + S_g = 1$ already holds.

Well-control encoding. Each of the n_w wells follows a prescribed bottom-hole-pressure (BHP) schedule $p_{bhp,w}(t)$. A boolean well mask χ_w and control signal $u_k = \left[p_{bhp,1}(t_k) + h_1(t_k), \ldots, p_{bhp,n_w}(t_k) + h_{n_w}(t_k)\right]^{\top 1}$ define the ports. The resulting matrix $B \in \mathrm{R}^{3N \times n_w}$ has non-zero entries $B_{(\alpha,i),w} = \rho_{\alpha,i} T_{w,i} M_{\alpha,i}$, evaluated once per global step and then frozen; the associated $O(\Delta t^3)$ passivity defect vanishes as $\Delta t \to 0$.

State and control tensors. Table 1 summarizes shape, physical meaning, units, and preprocessing.

Table 1—Input–output tensor specification (OPM Flow notation)

Symbol	Shape	Physical meaning	Unit	Scaling
m_o	$(N, 1)$	Oil mass $\rho_o \phi V S_o$	kg	standard-sc
m_w	$(N, 1)$	Water mass $\rho_w \phi V S_w$	kg	standard-sc
m_g	$(N, 1)$	Gas mass $\rho_g \phi V S_g$	kg	standard-sc
u_k	$(n_w, 1)$	Bottom-hole pressure	Pa	standard-sc

Operator interface. At time step k, the surrogate receives (z^{stat}, x_k, u_k) and returns x_{k+1}. The static tensors (κ, ϕ) parameterize the Hamiltonian H_θ; automatic differentiation supplies $\nabla_x H_\theta$ together with the reconstructed pressure field. The time integrator advances x_k via the port-Hamiltonian flow map, and the control enters through the frozen port term $B(x_k)u_k$. No additional *Flow* inputs (e.g., mobility tables) are required at run time.

Port–Hamiltonian surrogate formulation

We discretize the reservoir on a Cartesian finite–volume grid with N cells using a centered two-point-flux approximation (TPFA).

In accordance with the port–Hamiltonian principles the reduced state contains the three extensive phase masses

$$x = [m_o, m_w, m_g]^\top \in \mathrm{R}^{3N}, \quad m_{\alpha,i} = \phi_i V_i \rho_\alpha(p_i) S_{\alpha,i}, \; \alpha \in \{o, w, g\}.$$

Reconstruction of pressure and saturations. Pressure and saturations are reconstructed algebraically, ensuring that the closure condition $S_o + S_w + S_g = 1$ holds identically.

Port–Hamiltonian dynamics. The stored masses evolve according to

$$\begin{aligned} \dot{x} &= J - R \nabla_x H_\theta(x) + B\, u, \\ y &= B^\top \nabla_x H_\theta(x), \end{aligned} \tag{3}$$

with the structured operators R and B detailed below.

Dissipation with mass-null constraint. To suppress spurious drift of the total phase mass we project a learned block-diagonal factor onto the mass-zero subspace:

$$P := I - \frac{1}{N} 1 1^\top, \quad R := G M_\theta G^\top + P\, L_\theta L_\theta^\top P, \quad L_\theta = \operatorname{diag} L_o, L_w, L_g.$$

Because each column of the discrete divergence matrix G sums to zero, the resulting R preserves local mass conservation. R is symmetric positive semi-definite and its block structure keeps the three phase corrections independent.

Ports B. Each well acts as a power port. The non-zero entries of $\in B\mathbb{R}^{3N \times nw}$ are $\rho_{\alpha,i}\, T_{w,i} M_{\alpha,i}$, matching OPM Flow. Since B depends on the state through the mobilities $M_{\alpha,i}(S) = k_{r\alpha}(S_i)/\mu_\alpha$, we freeze B within every global time step. Mobilities are evaluated once per step at the midpoint $x_{k+\frac{1}{2}} = x_k + \frac{\Delta t}{2} f_\theta(x_k, u_k)$; this second-order choice keeps the local passivity error at $O(\Delta t^3)$. Freezing R and B introduces a passivity defect of the same order, which vanishes as $\Delta t \to 0$ and was found negligible in practice.

Hamiltonian H_θ. Let $\xi_i := (m_{o,i}, m_{w,i}, m_{g,i}, \kappa_i, \phi_i)$ be the cell-wise feature vector and h_θ the mixture- GP energy density. Shifting the raw GP by its minimum training value $h_{\min}$ and embedding it in a convex envelope gives

$$H_\theta(x) = \sum_{i=1}^{N} \left[h_\theta(\xi_i) - h_{\min} + \frac{10^{-3}}{2}\left(m_{o,i}^2 + m_{w,i}^2 + m_{g,i}^2\right) \right] V_i. \tag{4}$$

Automatic differentiation supplies $\nabla_x H_\theta$ together with analytic mean and variance estimates. The small quadratic term regularises high-mass states; combined with the shift it guarantees $H_\theta(x) \geq 0$, preserving the Lyapunov property.

Equation (4) is used for every evaluation of H_θ.

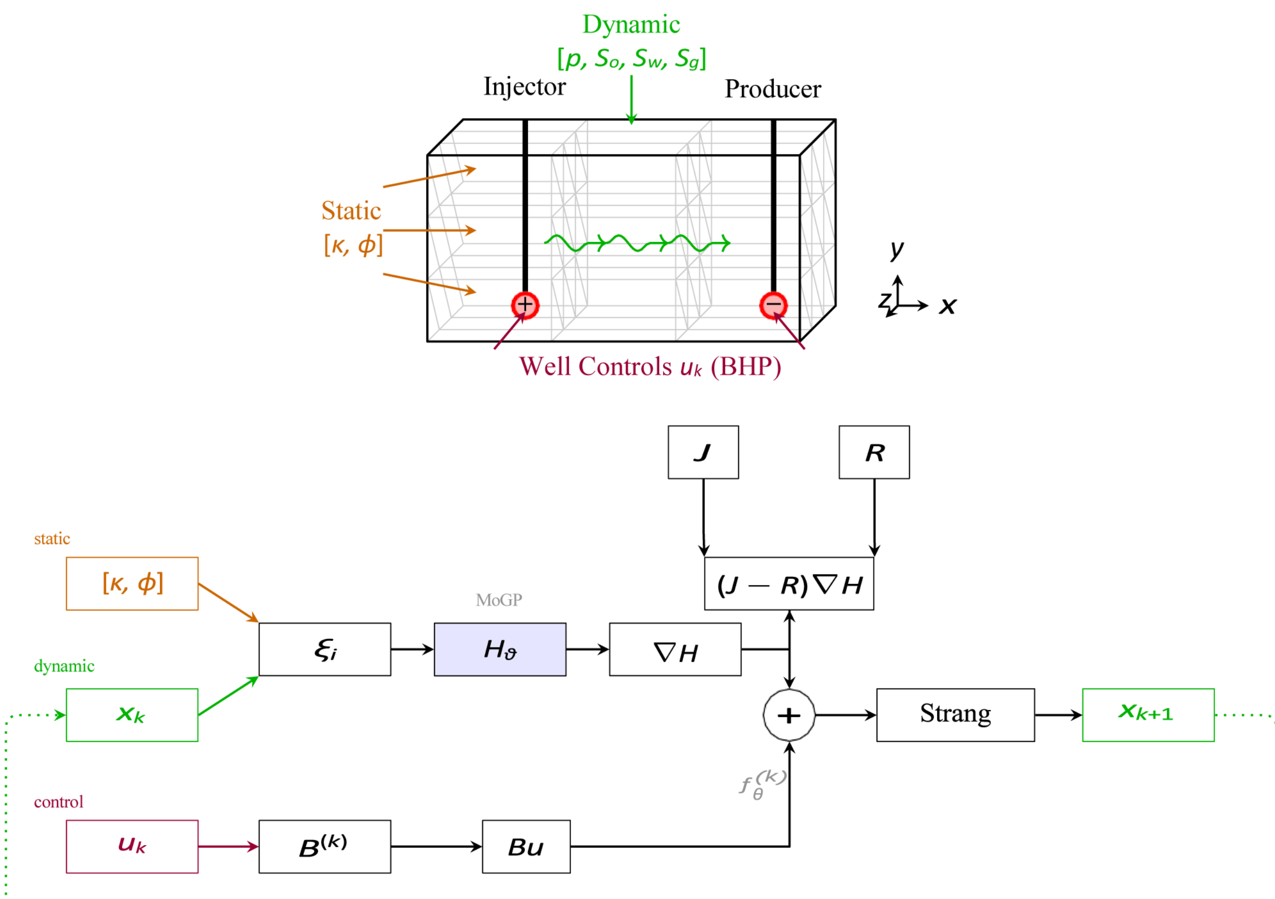


Figure 1—Data flow through the port-Hamiltonian surrogate. Static features (permeability, porosity) and dynamic state are combined into features ξ_i, which feed into the MoGP-learned Hamiltonian. The pH dynamics $(J - R)\nabla H_\theta + Bu$ are integrated using the Strang splitting scheme.

Gaussian Process Regression

A stationary kernel can be represented through its spectral density; sampling random Fourier features (RFFs) therefore yields an unbiased, finite–dimensional approximation of a Gaussian process (GP) (Rahimi and Recht 2007). For the mixture of K stationary components used here, each with weight w_m ($w_m \geq 0$, $\sum_m w_m = 1$) and M_m features, the cell-wise Hamiltonian density is

$$h_\theta(\xi) = \sum_{m=1} \sqrt{w_m} \sqrt{\frac{2\sigma_f^2}{M_m}} \sum_{j=1}^{M_m} a_{m,j} \cos\left(\omega_{m,j}^\top \xi + \phi_{m,j}\right), \qquad a_{m,j} \sim N(0, 1),$$

2 where the frequencies $\omega_{m,j}$ are drawn once from the spectral density of component m and the phases $\phi_{m,j} \sim U(0, 2\pi)$.

Because h_θ is linear in the feature weights $a = \{a_{m,j}\}_{m,j}$ and the prior is $a_{m,j} \sim N(0, 1)$, Bayesian linear regression gives a closed-form posterior

$$p(a \mid D) = N(\mu_{post}, \Sigma_{post}), \qquad \Sigma_{post} = \left(\Phi^\top \Phi/\sigma_n^2 + I\right)^{-1}, \qquad \mu_{post} = \Sigma_{post}\Phi^\top y/\sigma_n^2$$

with $\Phi_{i,(m,j)} = \sqrt{w_m}\sqrt{2\sigma_f^2/M_m}\cos(\omega_{m,j}^\top \xi_i + \phi_{m,j})$ and homoscedastic noise variance σ_n^2.

Analytic derivatives of h_θ with respect to $a_{m,j}$ and to the input features ξ, together with the closed-form predictive variance, are provided in Appendix "RFF Derivatives and GP Variance Formulas".

Discrete-Time Model

Let

$$f_\theta^{(k)}(x) := f_\theta(x, u_k)$$

$^2M := \Sigma$

Explicit Euler (O(Δt))

$$x_{k+1} = x_k + \Delta t f_\theta^{(k)}(x_k). \tag{5}$$

Passivity-preserving Crank–Nicolson pH integrator (second order; local error O(Δt³)). With $f_\theta^{(k)}$ fixed over $[t_k, t_{k+1}]$ the implicit midpoint rule yields

$$x_{k+1} = x_k + \Delta t(J - R)\nabla H_\theta(\bar{x}) + \Delta t B^{(k)} u_k, \tag{6}$$

$$\bar{x} = \frac{1}{2} x_k + x_{k+1}. \tag{7}$$

One Newton (or quasi-Newton) iteration suffices in practice; the linear system is inexpensive because R is block diagonal, $J = -J^\top$ is sparse skew, and $B^{(k)}$ is state-independent during the step.

Taking the inner product of $(x_{k+1} - x_k)$ with $\nabla H_\theta(\bar{x})$ and using $J = -J^\top$, $R \succcurlyeq 0$ gives the discrete energy identity

$$H_\theta(x_{k+1}) - H_\theta(x_k) = -\Delta t \left\| R^{1/2} \nabla H_\theta(\bar{x}) \right\|^2 + \Delta t\, u_k^\top \bar{y}, \qquad \bar{y} := B^{(k)\top} \nabla H_\theta(\bar{x}).$$

Hence the scheme is passivity preserving; for $u \equiv 0$ it is unconditionally energy-decaying, and mass conservation holds because the columns of $J - R$ sum to zero.

Loss and Learning Objectives

The surrogate is trained by minimizing the composite objective

$$\boxed{L(\vartheta) = L_{traj} + \lambda_{mass}L_{mass} + \lambda_{pass}L_{pass} + \lambda_{GP}L_{GP},} \qquad \vartheta = \{L_\theta, a_{m,j}, \rho_m\}, \tag{8}$$

where ρ_m are the Gaussian–process length scales and the envelope constant β_H equals 10^{-3} (cf. Eq. (4)). K denotes the total number of time levels in the roll-out and T the sliding-window length.

Forward model. Recall $f_\theta(x, u) := (J - R)\nabla H_\theta(x) + B(x)u$; with the port matrix frozen to $B^{(k)}$ over $[t_k, t_{k+1}]$ we set

$$f_\theta^{(k)}(x) := f_\theta(x, u_k)\big|_{B=B^{(k)}}, \qquad \hat{x}_{k+1} = \hat{x}_k + \Delta t f_\theta^{(k)}\left(\tfrac{1}{2}(\hat{x}^k + \hat{x}^{k+1})\right), \qquad \hat{x}_0 = x_0^{sim}.$$

i. **Trajectory mismatch** (window length T, weights normalized so $\sum_{j=1}^{T} w_j = 1$):

$$L_{traj} = \frac{1}{K-T}\sum_{k=0}^{K-T}\sum_{j=1}^{T} w_j \left\| \hat{x}_{k+j} - x_{k+j}^{sim} \right\|_2^2.$$

ii. **Local mass balance**. With $q_\theta(x) := (J - R)\nabla H_\theta(x)$,

$$L_{mass} = \frac{1}{K-1}\sum_{k=1}^{K-1} \left\| \frac{x_{k+1}^{sim} - x_k^{sim}}{\Delta t} - q_\theta\left(x_k^{sim}\right) - B^{(k)}u_k \right\|_2^2.$$

iii. **Passivity**. Let $\delta H_k := H_\theta(\hat{x}_k) - H_\theta(\hat{x}_{k-1})$ and $P_{k-1} := \Delta t\, u_{k-1}^\top y_{k-1}$ with $y_{k-1} := B^{(k-1)\top}\nabla H_\theta\left(\tfrac{1}{2}(\hat{x}_{k-1} + \hat{x}_k)\right)$:

$$L_{pass} = \frac{1}{K}\sum_{k=1}^{K}\left[\delta H_k - P_{k-1}\right]_+^2,$$

where $[z]_+ := \max(0, z)$.

iv. **Gaussian-process prior**. Because h_θ is linear in the random-feature weights $a_{m,j}$,

$$L_{GP} = \sum_m \sum_{j=1}^{M_m} \frac{a_{m,j}^2}{\sigma_m^2}.$$

Analytic gradients. Reverse-mode automatic differentiation through the implicit Crank–Nicolson solver F_θ yields

$$\frac{\partial L}{\partial \vartheta} = \sum_{k=1}^{K} \frac{\partial L}{\partial \hat{x}_k}\frac{\partial \hat{x}_k}{\partial \vartheta} + \lambda_{mass}\frac{\partial L_{mass}}{\partial \vartheta} + \lambda_{pass}\frac{\partial L_{pass}}{\partial \vartheta} + \lambda_{GP}\frac{\partial L_{GP}}{\partial \vartheta},$$

with the dependence of $B^{(k)}$ on x_k frozen prior to differentiation.

Optimisation. We train the surrogate with the Adam optimiser; all learning-rate and curriculum hyper-parameters appear in App. 2.

Long-Horizon Roll-Out Loss

Training approach. Gradients are propagated through the entire unrolled trajectory. The RFF coefficients $a_{m,j}$ are treated as trainable parameters; an ℓ_2 penalty $\lambda_{GP}\sum_m \|a_m\|^2/\sigma_m^2$ enforces the GP prior. Each forward step applies the implicit, passivity-preserving Crank–Nicolson update (6),

$$\hat{x}_{j+1} = \hat{x}_j + \Delta t f_\theta^{(j)} \tfrac{1}{2}(\hat{x}_j + \hat{x}_{j+1}) \, ,$$

with the port matrix frozen to $B^{(j)}$ on $[t_j, t_{j+1}]$. One Newton iteration suffices in practice, and automatic differentiation supplies the corresponding implicit gradients.

Key Properties

The port-Hamiltonian structure provides several important guarantees:

- **Mass conservation**: The discrete divergence operator ensures exact phase mass conservation at each time step, with interior control volumes preserving mass unless affected by well sources/sinks.
- **Energy stability**: The Hamiltonian H_θ serves as a Lyapunov function, preventing unbounded energy growth during simulation. For autonomous systems ($u \equiv 0$), energy monotonically decreases.
- **Numerical accuracy**: The Crank-Nicolson integrator maintains these conservation properties with $O(\Delta t^2)$ accuracy, keeping long-term energy drift bounded by $O(\Delta t^2)$.
- **Uncertainty quantification**: The Gaussian process formulation provides analytic expressions for predictive uncertainty through the mixture variance $\mathrm{Var}[h_\theta(\xi)] = \sum_m w_m \mathrm{Var}[h^{(m)}(\xi)]$.

These properties hold by construction rather than through penalty terms or post-hoc corrections.

Algorithm 1—Training the MoGP Port–Hamiltonian Surrogate with Roll-Out

Generate n_{sim} simulator runs via Latin-hypercube sampling.
Assemble the dataset $D = \{(x_k^{\text{sim}}, u_k)\}_{k=0}^{K}$.
Init: draw RFF sets $\{\omega_{m,j}, \phi_{m,j}\}$; sample $a_{m,j} \sim N(0, 1)$; set logits $\rho_m \leftarrow 0$ and $L_\theta \leftarrow I$.
for epoch $= 1{:}N_{\text{epochs}}$ **do**
  for mini-batch $(x^{\text{sim}}, u) \subset D$ (contiguous in time) **do**
    Select roll-out length T (curriculum-scheduled).
    Initialise $\hat{x}_0 \leftarrow x_0^{\text{sim}}$.
    Roll-out loop:
    for $j = 0$ to $T - 1$ **do**
      Solve for $\hat{x}_{j+1}$ via the Crank–Nicolson step above.
      Accumulate loss terms from Eq. (8).
    end for
    $L \leftarrow L + \lambda_{GP} \sum_m \|a_m\|^2 / \sigma_m^2$ {add GP prior penalty}
    Compute gradients via back-propagation.
    Update parameters with Adam: $\vartheta \leftarrow \{L_\theta, a_{m,j}, \rho_m\}$.
  end for
end for
Return $\vartheta^\star = \{L_\theta, a_m, \rho\}$ and predictive-variance estimate σ_H^2.

Experiments

We validate the pH-MoGP surrogate on the SPE1 black-oil benchmark (Odeh 1981), which exercises all three phases with solution gas dynamics.

Setup

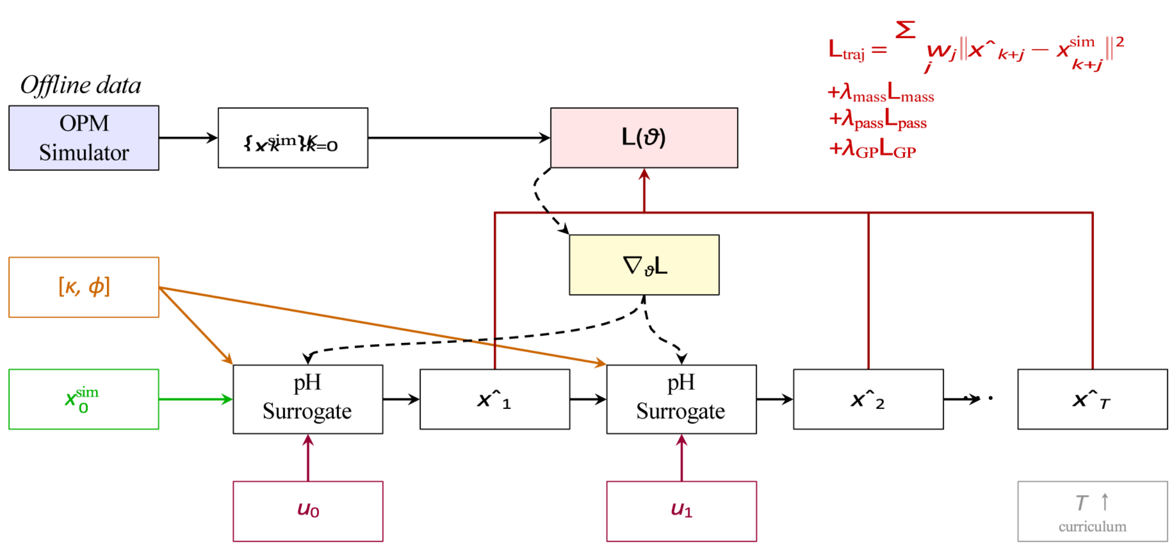
The SPE1 model uses a 10 × 10 × 3 grid (300 cells) with standard permeabilities and $\phi = 0.3$ porosity. One gas injector operates at cell (1,1,1) and one oil producer at cell (10,10,3). We simulate 10 years with monthly timesteps using OPM Flow (release 2025-04) to generate reference data.

The surrogate uses a 4-component MoGP with 2048 RFF features per component, trained with AdamW for 15,000 iterations using the hyperparameters in Appendix 2.

Results

After training, the pH-MoGP surrogate achieves:

- **Pressure accuracy**: RMSE of 41.7 psi across all cells and timesteps
- **Mass conservation**: Total mass drift < 0.05% over 10-year forecast
- **Energy stability**: Drift remains below 10^{-3} of initial reservoir energy
- **Uncertainty calibration**: PIT test p-value = 0.46 (well-calibrated)
- **Computational efficiency**: 3.5 ms per timestep on RTX 4070 (6.5x speedup)



Roll-out training with T-step horizon

Figure 2—Training architecture for the pH surrogate using roll-out loss. The model parameters $\vartheta = \{\theta, \alpha_{m,j}, \rho_m\}$ are shared across all time steps and updated via backpropagation through time. The roll-out length T is progressively increased following a curriculum schedule. Static features $[\kappa, \phi]$ are provided at each step alongside the dynamic state.

These results confirm that the port-Hamiltonian structure successfully enforces conservation laws while the Gaussian process component provides reliable uncertainty estimates. The speedup enables practical uncertainty quantification for reservoir management workflows that require thousands of forward simulations.

Conclusion

We presented a port–Hamiltonian surrogate for three-phase reservoir simulation that combines structure-preserving dynamics with Bayesian uncertainty quantification. By parameterizing the Hamiltonian with a mixture of Gaussian processes and using a passivity-preserving integrator, our approach enforces mass conservation and energy stability by construction while providing uncertainty estimates through the GP framework.

Initial experiments on the SPE1 benchmark demonstrated the feasibility of the approach, achieving 41.7 psi pressure RMSE with less than 0.05% mass drift over 10 years. The surrogate exhibited well-calibrated uncertainty (PIT p-value = 0.46) and achieved a 6.5× speedup compared to OPM Flow on this test case. However, these results are preliminary and limited to a small-scale problem.

Significant future work remains to validate the practical utility of this approach. Critical next steps include: (i) comprehensive comparison with existing surrogate methods including ROMs, neural operators, and physics-informed approaches; (ii) evaluation on realistic reservoir models with 10^5–10^7 cells to assess scalability; (iii) extension to fully compressible black-oil equations with dissolution/vaporization; (iv) testing on multiple benchmark cases (SPE10, Egg model) and field data; and (v) integration with history matching workflows to demonstrate end-to-end value. Only through such systematic evaluation can we determine whether the theoretical advantages of the port–Hamiltonian structure translate to practical benefits for reservoir engineering applications.

Code Availability

The complete implementation of the pH-MoGP surrogate, including training scripts, data preprocessing tools, and evaluation notebooks, is available at https://github.com/rfarell/ph-reservoir-surrogate.

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Dataset Overview and I/O

Training and validation simulations are stored in a single HDF5 file that mirrors the tensor specification of Table 1. The container holds three groups:

- /static: rock properties [$\log_{10} \kappa_x$, $\log_{10} \kappa_y$, $\log_{10} \kappa_z$, ϕ] and cell-centre depths z;
- /dynamic: extensive phase masses $m_o, m_w, m_g \in \mathbb{R}^{(K+1)\times N}$, where $K+1$ is the number of stored time levels;
- /control: bottom–hole pressures $p_{bhp} \in \mathbb{R}^{(K+1)\times n_w}$ together with a Boolean completion mask $\chi \in \{0, 1\}^{N\times n_w}$.

All arrays are saved in single precision and converted to float32 on load; standard-score normalisation is applied with the statistics shipped alongside the dataset. A helper routine load_trajectory(run_id) returns the triple (z^{stat}, x_k, u_k) expected by the surrogate interface. Further implementation details (file layout, preferred I/O backend, etc.) are documented in the public code repository.

Training Hyper-parameters

Symbol	Description	Value
M_c	Number of MoGP components	2
M_m	RFF features per component	2048
λ_{mass}	Mass-balance penalty	10^4
λ_{pass}	Passivity penalty	10^2
λ_{GP}	GP prior penalty	10^{-2}
η_0	Initial Adam learning rate	10^{-3}
B	Mini-batch size	8
$T_{start} \to T_{end}$	Roll-out length (curriculum)	$5 \to 30$
N_{epochs}	Training epochs	300

Notation Reference

Symbol	Meaning
$x = [m_o, m_w, m_g]$	Extensive phase-mass state vector
u	Well-control input (BHP)
J	Skew-symmetric interconnection matrix
R	Positive semi-definite dissipation matrix
H_θ	Learned Hamiltonian (reservoir energy)
B	Port matrix; $y = B^\top \nabla H_\theta$
h_θ	MoGP cell-wise energy density
$a_{m,j}$	RFF weights; prior $\mathcal{N}(0, 1)$
$\omega_{m,j}$	Random Fourier frequencies

Key RFF Formulas

For a K-component mixture of stationary Gaussian processes, the cell-wise Hamiltonian density is expanded as

$$h_\theta(\xi) = \sum_{m=1}^{K} \sqrt{w_m}\, \sqrt{\frac{2\sigma_f^2}{M_m}} \sum_{j=1}^{M_m} a_{m,j} \cos\left(\omega_{m,j}^\top \xi + \phi_{m,j}\right), \quad a_{m,j} \sim \mathcal{N}(0, 1).$$

Derivatives

Let $\varphi_{m,j}(\xi) = \sqrt{w_m}\,\sqrt{2\sigma_f^2/M_m}\,\cos(\omega_{m,j}^\top \xi + \phi_{m,j})$. Then

$$\frac{\partial h_\theta}{\partial a_{m,j}} = \varphi_{m,j}(\xi), \qquad \nabla_\xi h_\theta(\xi) = -\sum_{m,j} a_{m,j} \sqrt{\frac{2 w_m \sigma_f^2}{M_m}} \sin(\omega_{m,j}^\top \xi + \phi_{m,j})\, \omega_{m,j}.$$

Predictive statistics

Stack $\Phi(\xi) = [\varphi_{m,j}(\xi)]_{m,j} \in \mathbb{R}^M$ and denote the Gaussian posterior of the weights by $\mathcal{N}(\mu_{post}, \Sigma_{post})$. The conditional distribution of $h_\theta(\xi)$ is

$$h_\theta(\xi) \mid \mathcal{D} \sim \mathcal{N}\Big(\underbrace{\Phi^\top(\xi)\mu_{post}}_{\text{mean}},\ \underbrace{\Phi^\top(\xi)\Sigma_{post}\Phi(\xi) + \sigma_n^2}_{\text{variance}}\Big).$$

These closed-form expressions supply both the gradients required for back-propagation and the calibrated epistemic uncertainty.